Geometry
Unit 8
Lesson 2 Homework

Name Answer Key ___________
Date____________________
Period___________________

Use the given information to answer questions 1-3.

Quadrilateral $ADHP$ is shown with $HP = 25$, $AD = 8x + 21$, $DH = 20x - 4$ where $x = \frac{1}{2}$.

1. I can conclude …

   that quadrilateral $ADHP$ is a parallelogram.

2. I know this because …

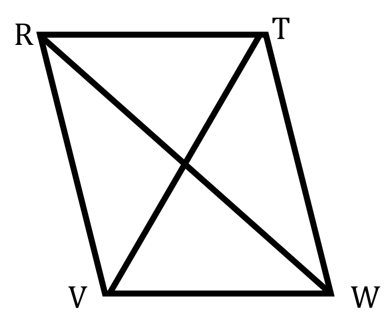
   there are two pairs of opposite sides of equal length.

3. If $ADHP$ is a parallelogram, what is the length of $\overline{PA}$?

   $PA = 6$

Use the given information to answer questions 4-6.

Quadrilateral $RTWV$ is shown with diagonals $\overline{RW}$ and $\overline{TV}$ that intersect at point M where $RM = 4x$, $TM = (3y + 15)$, $WM = 84$, and $VM = 75$.



4. What value of x will make $RTWV$ a parallelogram?

   $4x = 84$
   $x = 21$

5. What value of y will make $RTWV$ a parallelogram?

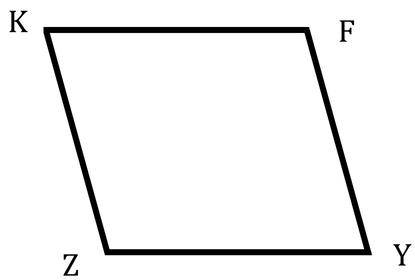
$3y + 15 = 75$
$3y = 60$
$y = 20$

6. What is the length of $\overline{VT}$?

150

Use the given information to answer questions 7-10.

Quadrilateral $KFYZ$ is shown where $m\angle F = (20z - 4)°$, $m\angle Y = 54°$, and $m\angle Z = (8t)°$, where $t = 15\frac{3}{4}$.



Devonte and Briyel are discussing whether $KFYZ$ is a parallelogram. Briyel says it is, and Devonte says it is not.

7. Who is correct?

Briyel is correct.

8. Which property of parallelograms was used to determine if $KFYZ$ is a parallelogram?

Consecutive interior angles are supplementary.

9. What is the value of z?

$20z - 4 = 126$
$20z = 130$
$z = 6.5$

10. What is the $m\angle K$?

$m\angle K = 54°$